FINAL PROJECT

Movies Recommendation system
— Data & Model results

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01

Idea & Dataset

Where the numbers come from, and how we made them trustworthy

FROM RAW DATA TO OUR DATASET

From almost 400K movies down to about 50K.

~400K

Original dataset

TMDB + IMDb movies

→

69,439

After filtering

Over 10 votes, status Released, no missing title, overview, genres, cast, directors or release date

→

~50K

Stratified sample

Balanced by primary genre and decade, so classic movies are not neglected

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The final sample holds 49,994 movies when loaded in the insights notebook.

THE DATASET

TMDB movies merged with IMDb ratings.

49,994
movies loaded, 29 columns

49,928
after removing 66 duplicates

13,318
movies with 100+ votes (27%)

Budget and revenue are recorded for only about 20% of movies: 10,168 usable budgets and 9,563 usable revenues.

DATA PREPARATION

Four steps to make the insights trustworthy.

1 · De-duplicate

66 repeated IMDb/TMDB ids removed early.

2 · Clean the zeros

About 80% of budget and revenue were zeros meaning "not recorded". They are ignored, along with 179 budgets and 115 revenues under $1,000.

3 · Reliability filter

Rating insights use only movies with at least 100 votes.

4 · Corrected heatmap

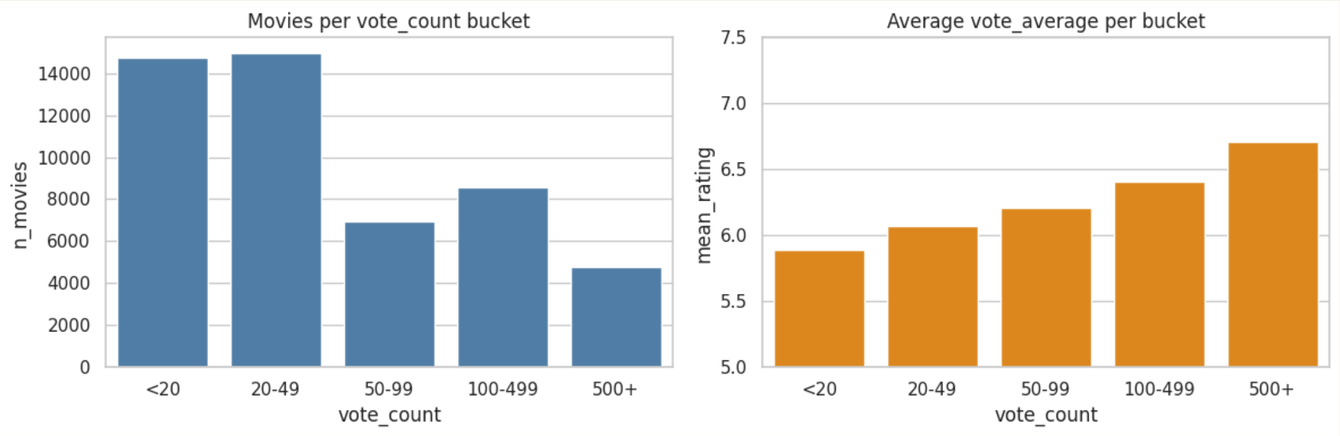
Correlations recomputed without the fake zeros.

02

THE DISCOVERIES

Ratings · Money · People · Content

Not every rating deserves the same trust.



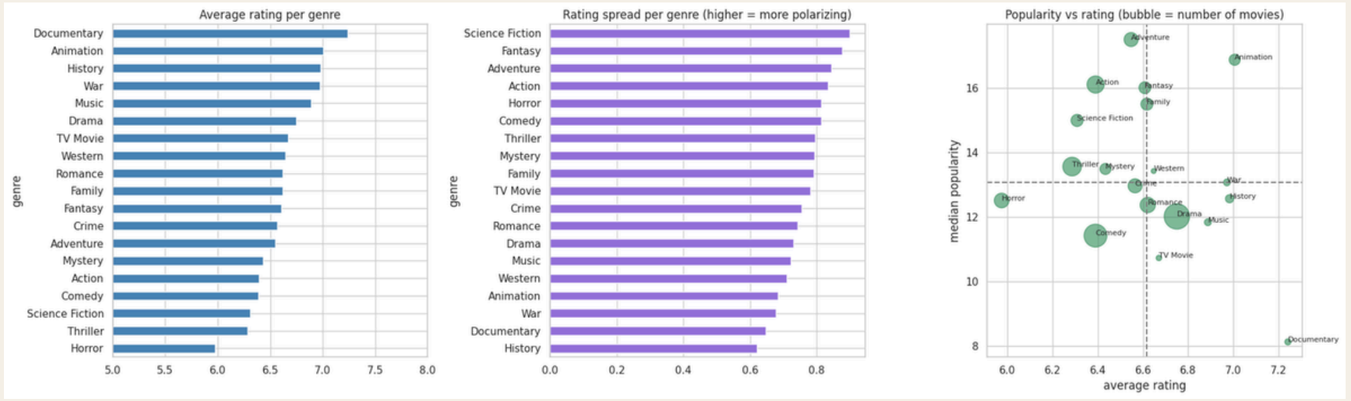
59%

of movies have fewer than 50 votes

Average rating rises from 5.89 (under 20 votes) to 6.71 (500+ votes), so we keep movies with 100+ votes.

A · RATINGS & QUALITY

Popularity and quality don't always tell the same story.



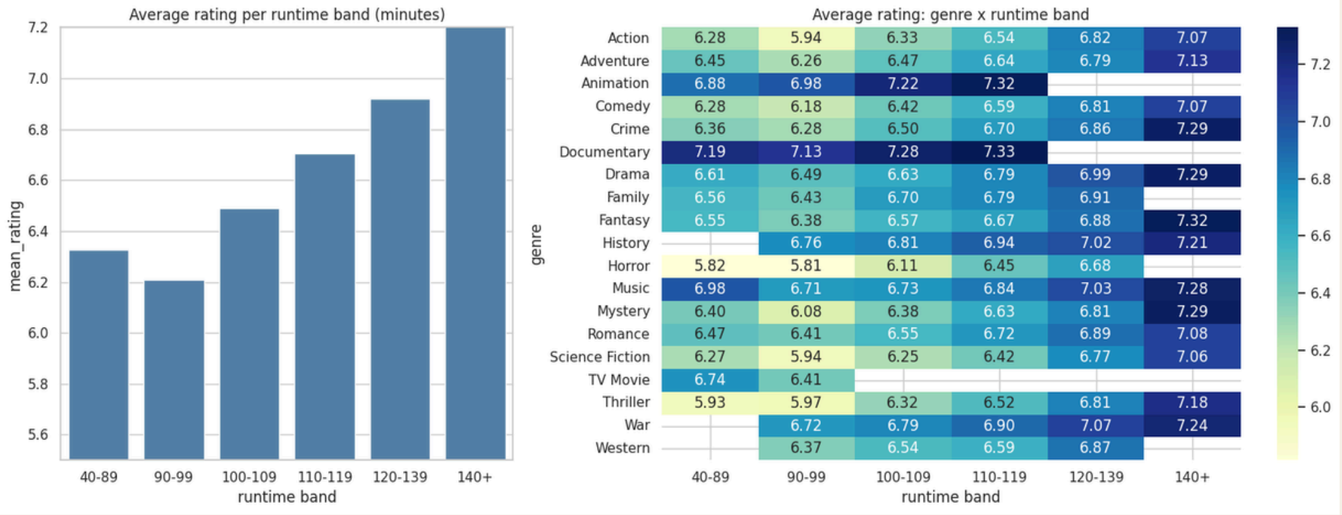
7.24

Documentary: best rated, yet the least popular genre

Animation is the only genre that is both highly rated (7.01) and very popular. Horror is lowest at 5.97.

A · RATINGS & QUALITY

Longer movies are rated higher.



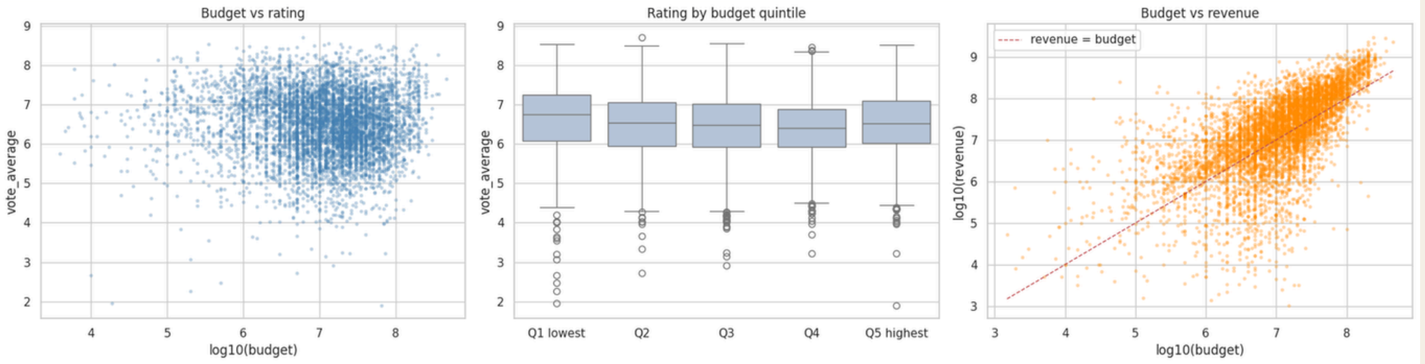
6.21 → 7.23

from 90–99 minutes to 140+ minutes

Correlation 0.30. Likely a selection effect: long movies tend to be ambitious projects and classics.

B · MONEY

Does spending more actually translate into more money?



0.65

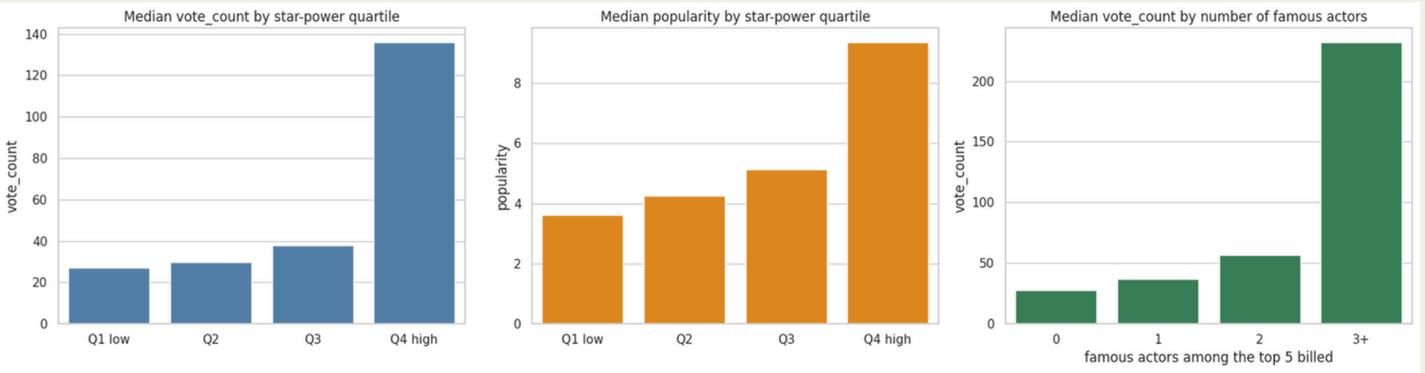
budget vs revenue (log correlation)

Budget vs rating: **-0.06**. Money buys revenue, not quality.

66.5% of movies earn more than their budget, a median of 1.84× (not profit; inflation is not adjusted).

C · PEOPLE

Famous casts bring attention, not better ratings.



136 vs 27

median votes: top vs bottom star-power quartile

The average rating stays flat (6.19 → 6.24). This is association, not proof that stars cause the extra votes.

D · CONTENT & AUDIENCE

Hidden gems: loved, but little known.

442
hidden gems (7.5+ on TMDB and IMDb)

322
after excluding Music, Documentary, TV Movie

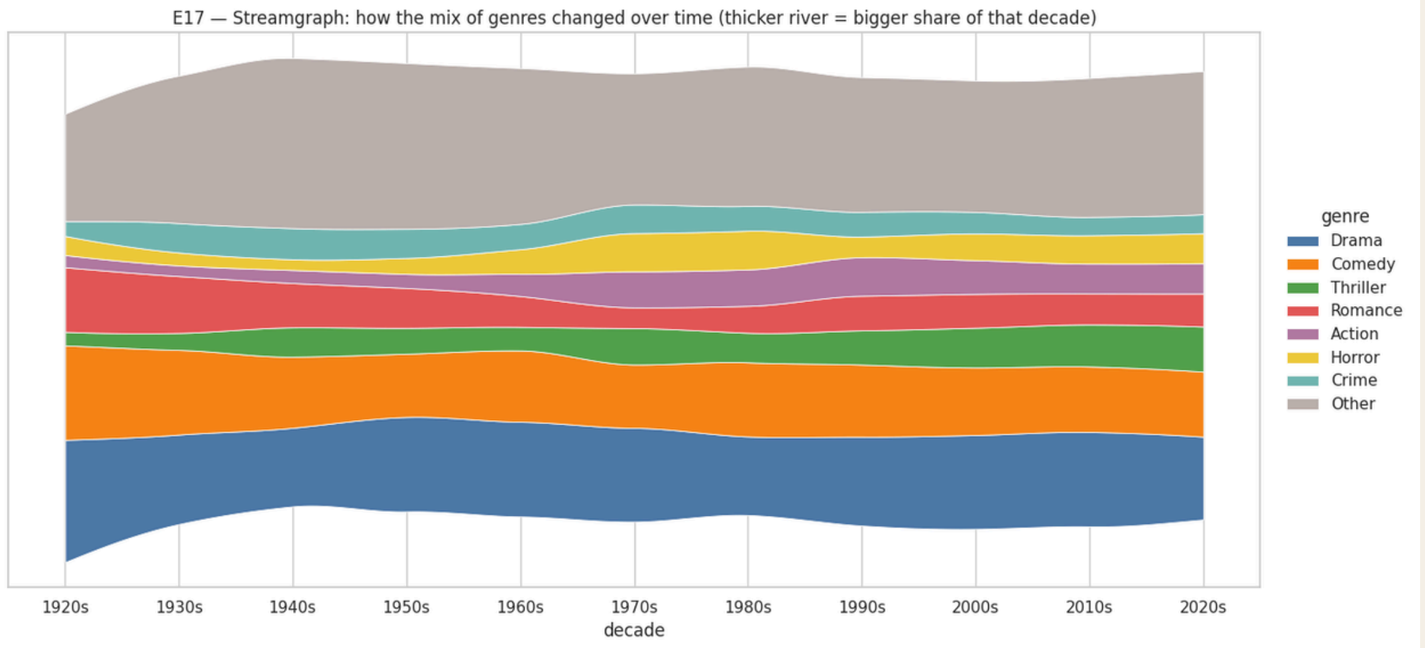
Rule: 100+ votes and popularity below the median of 12.4. Examples: Doctor Who: The Day of the Doctor, Apur Sansar, Samurai Rebellion.

03

GENRE CHEMISTRY

Which mixes work, and how genres changed over time

Romance fades while Thriller grows.

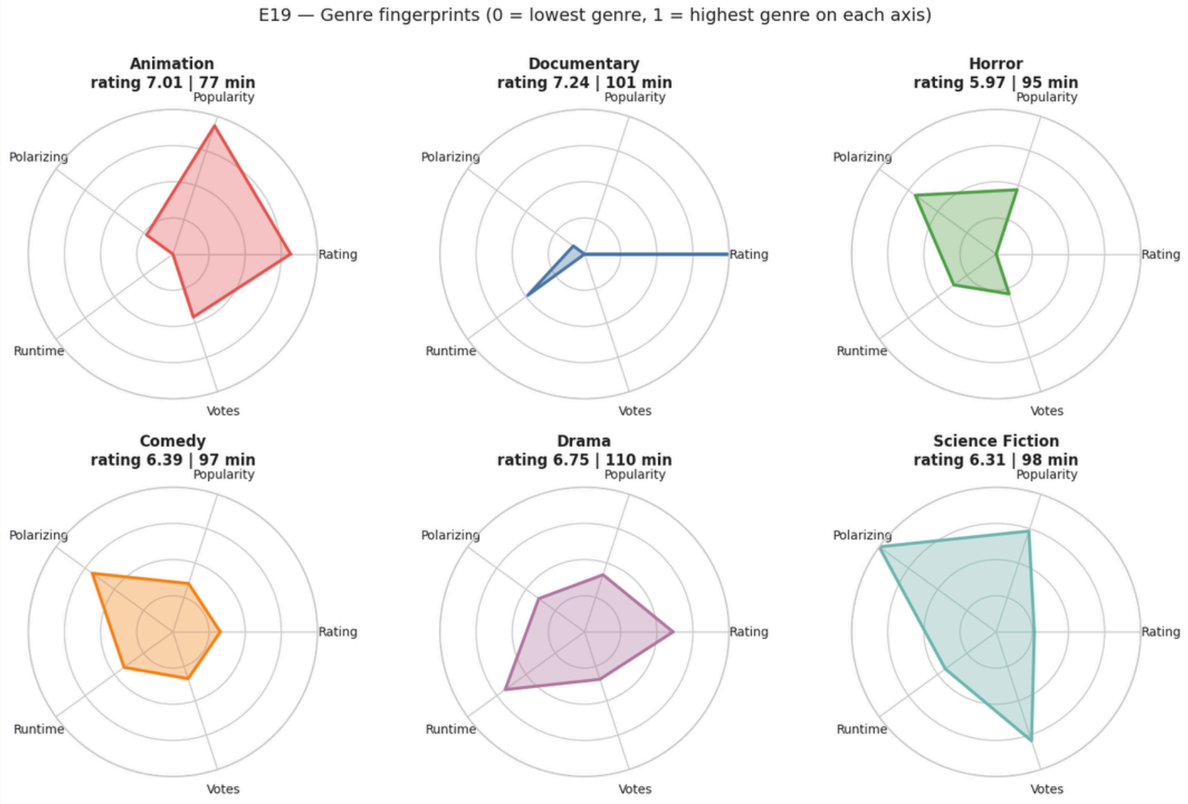


3% → 10%

Thriller share of genre tags, 1920s to 2020s

Romance falls from 14.4% to 7.4%. Drama stays the biggest genre in every decade.

Every genre has its own personality.

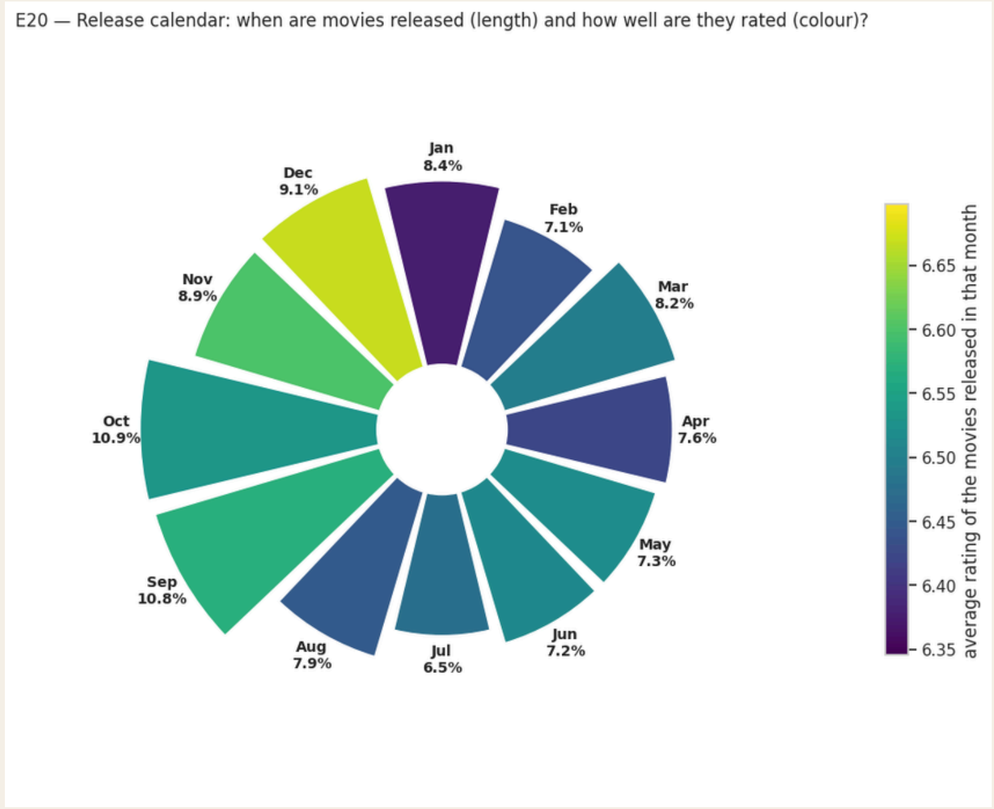


0.90

Science Fiction: the most polarizing genre

Animation is the short (77 min) crowd-pleaser; Documentary has the top rating but the fewest votes.

Autumn is busy, and year-end releases rate highest.



21.7%

of releases come in September and October

November and December average 6.60 and 6.67. January is lowest at 6.38. A pattern, not proof.

04

UNDER THE HOOD

Preprocessing, embeddings and models

PREPROCESSING FOR THE MODELS

From raw columns to a clean modeling table.

Drop and simplify

Removed ids, image paths and the constant status column; kept "has homepage" as a flag; adult titles removed.

Fill missing values

Languages from original language, countries from production companies, and keywords from the 15 most similar movies (12,960 rows).

Create features

Tagline flag and length, release day, month and year, and a combined text field.

Zeros become missing

Budget, revenue and runtime of 0 are treated as unknown, not as real values.

EMBEDDING TECHNIQUES

Eight blocks describe every movie.

Block	Technique	Weight
Text	SBERT (all-MiniLM-L6-v2) on title, tagline, overview, keywords	37%
Genres	Binary multi-hot vector	15%
Keywords	TF-IDF + SVD (64 dimensions)	15%
Cast (top 5)	TF-IDF + SVD (64 dimensions)	10%
Directors	TF-IDF + SVD (32 dimensions)	10%
Companies · Writers	TF-IDF + SVD (32 dimensions)	5% · 3%
Meta	Year, log runtime, log budget, language (standardized)	5%

FUSING THE BLOCKS

One weighted vector per movie, compared by cosine similarity.

651
dimensions in the fused embedding

49,853
movies embedded

Guessed keywords and very short descriptions (97 movies) count at half weight. Recommendations average the liked movies into one profile and return the nearest ones.

MODELS APPLIED

Learning which signals predict a good recommendation.

XGBoost
Learns non-linear combinations of the signals.

KNN
Pairs that look alike were usually the same creative team.

Logistic regression
A simple linear baseline.

Stacked model
Combines the three, with folds grouped by seed movie.

Weak label: same director or top-3 actor AND a shared genre. 120,000 seed-candidate pairs, 3.9% positive, 833 held-out test movies.

WHAT XGBOOST RELIES ON

Text and production companies matter most.

25% text similarity

24% production companies

13% same language

10% meta block

Cast, directors and writers were hidden from the models, because they define the label.

EVALUATION ON HELD-OUT MOVIES

The learned re-rankers beat random, and the hybrid wins on team hits.

Method	Team and genre hit @10	Genre overlap @10
Random order	0.036	0.675
Cosine (no people blocks)	0.061	0.718
XGBoost	0.075	0.673
Stack (XGB + KNN + LR)	0.074	0.673
Blend 70% cosine + 30% stack	0.064	0.714
Hybrid E (uses cast and directors)	0.161	0.675

Hybrid E is favored on team hits because it uses people. Genre overlap was not used for training, so it is the fairer check.

THE FINAL RECOMMENDER

One function, many ways to search.

Filters
Actor, director, genre, language or year decide which movies are allowed.

Profile
Liked movies, a free-text description, or only filters.

Hybrid search + models
Nearest movies by cosine, re-ranked by the stacked model.

Quality and diversity
A small push for well-rated movies and optional diversity re-ranking.

Duplicates and compilations are always excluded.

05

THE BIGGER PICTURE

What the data taught us

WHAT THE DATA TAUGHT US

Five lessons from 50,000 movies.

- Money buys revenue (0.65), not quality (-0.06).
- Famous casts and taglines bring about 2–5× the attention, with no rating gain.
- Popular and highly rated are different things: only Animation is both.
- Longer, older and heritage-themed movies score higher, partly through selection effects.
- Mixing Horror pulls ratings down; Animation mixes lift them.

READ WITH CARE

What this analysis cannot say.

27%
of movies have 100+ votes

~20%
have recorded budget or revenue

Everything here is association, not causation. Money is not inflation-adjusted. Vote counts, popularity and revenue are outcomes, so they must be excluded if ratings are later predicted.

Ratings, money, people and genres each tell a different story.

Thank you · Questions?